```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.metrics import (
    confusion_matrix,
    accuracy_score,
    precision_score,
    recall_score,
    f1_score,
    classification_report
)

print("Libraries imported successfully!")
```

```
Libraries imported successfully!
```

```
# Load Telco Customer Churn dataset

url = "https://raw.githubusercontent.com/IBM/telco-customer-churn-on-icp4d/

df = pd.read_csv(url)

print("Dataset loaded successfully!")
print("Dataset shape:", df.shape)

df.head()
```

```
Dataset loaded successfully!
Dataset shape: (7043, 21)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneServ
0	7590-VHVEG	Female	0	Yes	No	1	
1	5575-GNVDE	Male	0	No	No	34	
2	3668-QPYBK	Male	0	No	No	2	
3	7795-CFOCW	Male	0	No	No	45	
4	9237-HQITU	Female	0	No	No	2	

5 rows × 21 columns

```
print("Dataset Information:\n")
df.info()

print("\nMissing Values:\n")
print(df.isnull().sum())

print("\nDuplicate Rows:", df.duplicated().sum())
```

```
Dataset Information:

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB

Missing Values:

customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
```

```
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64

Duplicate Rows: 0
```

```
df.describe(include="all").T
```

	count	unique	top	freq	mean	std	min
customerID	7043	7043	3186-AJIEK	1	NaN	NaN	NaN
gender	7043	2	Male	3555	NaN	NaN	NaN
SeniorCitizen	7043.0	NaN	NaN	NaN	0.162147	0.368612	0.0
Partner	7043	2	No	3641	NaN	NaN	NaN
Dependents	7043	2	No	4933	NaN	NaN	NaN
tenure	7043.0	NaN	NaN	NaN	32.371149	24.559481	0.0
PhoneService	7043	2	Yes	6361	NaN	NaN	NaN
MultipleLines	7043	3	No	3390	NaN	NaN	NaN
InternetService	7043	3	Fiber optic	3096	NaN	NaN	NaN
OnlineSecurity	7043	3	No	3498	NaN	NaN	NaN
OnlineBackup	7043	3	No	3088	NaN	NaN	NaN
DeviceProtection	7043	3	No	3095	NaN	NaN	NaN
TechSupport	7043	3	No	3473	NaN	NaN	NaN
StreamingTV	7043	3	No	2810	NaN	NaN	NaN
StreamingMovies	7043	3	No	2785	NaN	NaN	NaN
Contract	7043	3	Month-to-month	3875	NaN	NaN	NaN
PaperlessBilling	7043	2	Yes	4171	NaN	NaN	NaN
PaymentMethod	7043	4	Electronic check	2365	NaN	NaN	NaN
MonthlyCharges	7043.0	NaN	NaN	NaN	64.761692	30.090047	18.25

```
df["TotalCharges"] = pd.to_numeric(
    df["TotalCharges"],
    errors="coerce"
)

# Fill missing TotalCharges with median
```

```
df["TotalCharges"] = df["TotalCharges"].fillna(
    df["TotalCharges"].median()
)

print("Missing values after cleaning:")
print(df.isnull().sum())
```

```
Missing values after cleaning:
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

```
# CustomerID does not help predict churn
df = df.drop("customerID", axis=1)

print("Customer ID removed.")
print("New shape:", df.shape)
```

```
Customer ID removed.
New shape: (7043, 20)
```

```
# Convert target column to numerical form
df["Churn"] = df["Churn"].map({
    "No": 0,
    "Yes": 1
})

# Convert all remaining categorical columns using one-hot encoding
df_encoded = pd.get_dummies(
    df,
    drop_first=True
)

print("Encoding completed!")
print("Encoded dataset shape:", df_encoded.shape)
```

```
df_encoded.head()
```

```
Encoding completed!
Encoded dataset shape: (7043, 31)
```

	SeniorCitizen	tenure	MonthlyCharges	TotalCharges	Churn	gender_Male
0	0	1	29.85	29.85	0	False
1	0	34	56.95	1889.50	0	True
2	0	2	53.85	108.15	1	True
3	0	45	42.30	1840.75	0	True
4	0	2	70.70	151.65	1	False

5 rows × 31 columns

```
X = df_encoded.drop("Churn", axis=1)
y = df_encoded["Churn"]

print("Features shape:", X.shape)
print("Target shape:", y.shape)
```

Show hidden output

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42,
    stratify=y
)

print("Training samples:", X_train.shape[0])
print("Testing samples :", X_test.shape[0])
```

Show hidden output

```
model = DecisionTreeClassifier(
    criterion="gini",
    max_depth=5,
    random_state=42
)

model.fit(X_train, y_train)

print("Decision Tree trained successfully!")
```

Show hidden output

```
y_pred = model.predict(X_test)

print("Predictions generated successfully!")

print("\nFirst 20 predictions:")
print(y_pred[:20])
```

Show hidden output

```
accuracy = accuracy_score(
    y_test,
    y_pred
)

print("Accuracy:", round(accuracy, 4))
```

Show hidden output

```
precision = precision_score(
    y_test,
    y_pred
)

recall = recall_score(
    y_test,
    y_pred
)

f1 = f1_score(
    y_test,
    y_pred
)

print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-Score :", round(f1, 4))
```

Show hidden output

```
print("Classification Report:\n")

print(
    classification_report(
        y_test,
        y_pred,
        target_names=["No Churn", "Churn"]
    )
)
```

Show hidden output

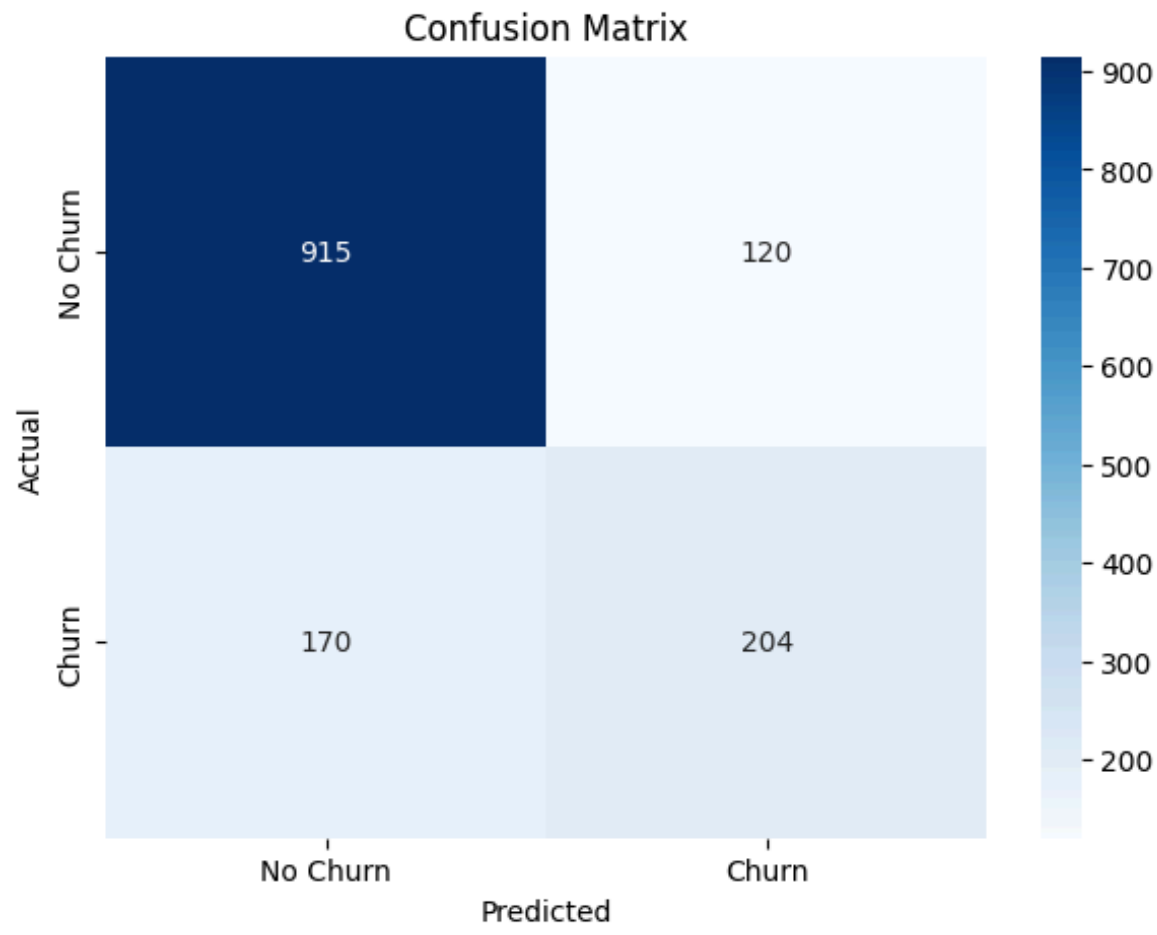
```
cm = confusion_matrix(
    y_test,
    y_pred
)

plt.figure(figsize=(7, 5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Churn", "Churn"],
    yticklabels=["No Churn", "Churn"]
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")

plt.show()
```

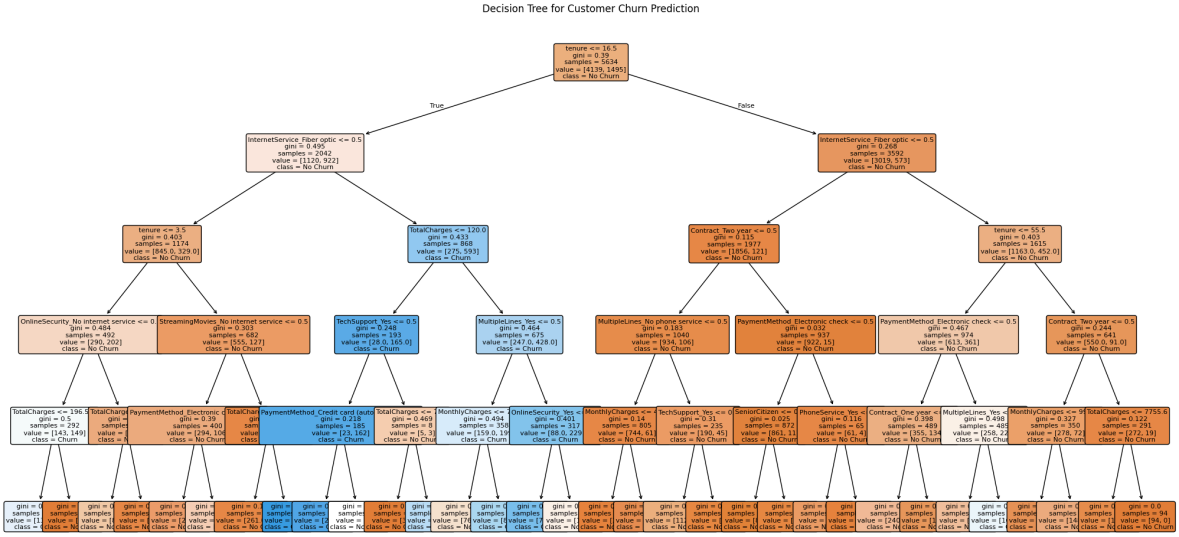



```
plt.figure(figsize=(25, 12))

plot_tree(
    model,
    feature_names=X.columns,
    class_names=["No Churn", "Churn"],
```

```
    filled=True,
    rounded=True,
    fontsize=8
)

plt.title("Decision Tree for Customer Churn Prediction")
plt.show()
```



```
feature_importance = pd.DataFrame({
    "Feature": X.columns,
    "Importance": model.feature_importances_
})

feature_importance = feature_importance.sort_values(
    by="Importance",
    ascending=False
)

feature_importance.head(15)
```

	Feature	Importance
1	tenure	0.421435
10	InternetService_Fiber optic	0.357542
3	TotalCharges	0.047129
28	PaymentMethod_Electronic check	0.036761
9	MultipleLines_Yes	0.023630
12	OnlineSecurity_No internet service	0.020669
25	Contract_Two year	0.019599
22	StreamingMovies_No internet service	0.017379
2	MonthlyCharges	0.016847
19	TechSupport_Yes	0.012988
24	Contract_One year	0.008553
8	MultipleLines_No phone service	0.007049
13	OnlineSecurity_Yes	0.006628
27	PaymentMethod_Credit card (automatic)	0.002533
7	PhoneService_Yes	0.000975

Next steps: Generate code with `feature_importance` New interactive sheet

```
top_features = feature_importance.head(10)

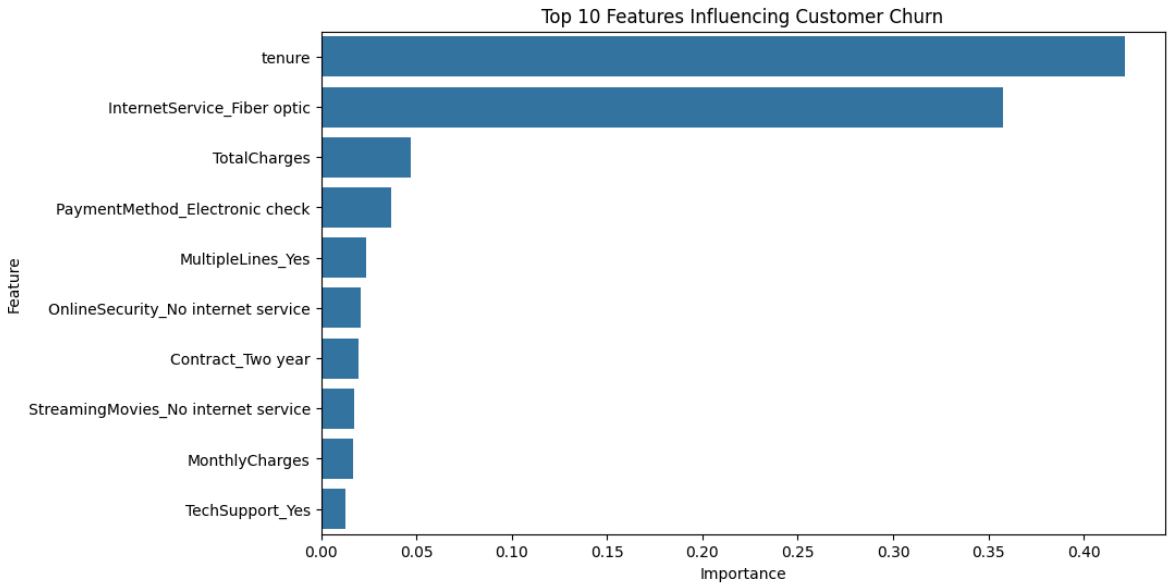
plt.figure(figsize=(10, 6))

sns.barplot(
    data=top_features,
    x="Importance",
    y="Feature"
)

plt.title("Top 10 Features Influencing Customer Churn")
```

```
plt.xlabel("Importance")
plt.ylabel("Feature")

plt.show()
```




```
comparison = pd.DataFrame({
    "Actual Churn": y_test.values,
    "Predicted Churn": y_pred
})

comparison.head(20)
```

	Actual Churn	Predicted Churn
0	0	0
1	0	1
2	0	0
3	0	0
4	0	0
5	0	0
6	0	0
7	0	0
8	0	0
9	1	0
10	0	0
11	0	0
12	0	0
13	1	1
14	0	0
15	0	0
16	0	0
17	0	1
18	0	0
19	0	0

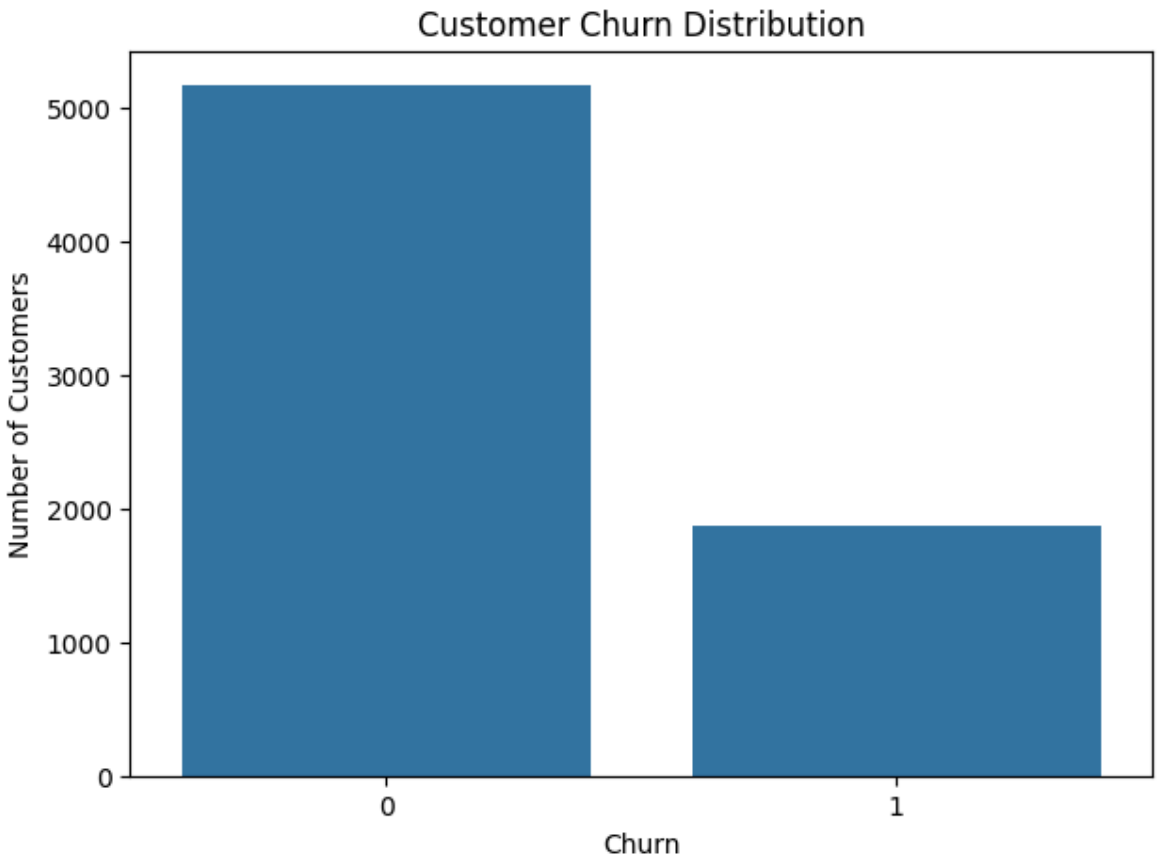
Next steps: Generate code with comparison | New interactive sheet

```
plt.figure(figsize=(7, 5))

sns.countplot(
    data=df,
    x="Churn"
)

plt.title("Customer Churn Distribution")
plt.xlabel("Churn")
plt.ylabel("Number of Customers")

plt.show()
```




```
print("========== MODEL PERFORMANCE ==========")

print("Accuracy :", round(accuracy, 4))
print("Precision:", round(precision, 4))
print("Recall   :", round(recall, 4))
print("F1-Score :", round(f1, 4))

print("\n========== TOP CHURN FACTORS ==========")

print(
    feature_importance.head(10).to_string(index=False)
)
```

```
========== MODEL PERFORMANCE ==========
Accuracy : 0.7942
Precision: 0.6296
Recall   : 0.5455
F1-Score : 0.5845

========== TOP CHURN FACTORS ==========
                           Feature  Importance
                            tenure    0.421435
        InternetService_Fiber optic    0.357542
                      TotalCharges    0.047129
     PaymentMethod_Electronic check    0.036761
                MultipleLines_Yes    0.023630
 OnlineSecurity_No internet service    0.020669
                Contract_Two year    0.019599
StreamingMovies_No internet service    0.017379
```

```
                    MonthlyCharges    0.016847
                   TechSupport_Yes    0.012988
```

```
print("""
CONCLUSION

A Decision Tree Classification model was successfully implemented
for customer churn prediction.

The dataset was preprocessed, categorical variables were encoded,
and the data was divided into training and testing sets using an
80:20 ratio.

The Decision Tree model was trained and evaluated using Accuracy,
Precision, Recall, F1-Score, and a Confusion Matrix.

The Decision Tree visualization and feature importance analysis
helped identify the major factors influencing customer churn.

These insights can help businesses identify customers at risk of
leaving and develop targeted retention strategies.
""")
```

```
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leaving and develop targeted retention strategies.
```

Ques 1: What is classification, and how does it differ from regression?

Ans: Classification is a supervised machine learning technique used to predict a categorical output or class. For example, in customer churn prediction, the model predicts whether a customer will churn or not churn. Regression is also a supervised learning technique, but it is used to predict continuous numerical values such as salary, house price, or sales. Therefore, classification predicts categories, while regression predicts numerical values.

Ques 2: Explain the working principle of the Decision Tree algorithm.

Ans: A Decision Tree is a supervised machine learning algorithm that makes predictions by dividing data into smaller groups using a series of decision rules. At each node, the algorithm selects a feature and a splitting condition that best separates the classes. This process continues until the data reaches a leaf node, where the final prediction is made. In customer churn prediction, the tree may make decisions based on factors such as contract type, tenure, monthly charges, and other customer characteristics.

Ques 3: Differentiate between Gini Index and Entropy as splitting criteria.

Ans: Gini Index and Entropy are measures used by Decision Trees to select the best feature for splitting the data. The Gini Index measures the impurity of a node based on the probability of incorrectly classifying a randomly chosen data point. Entropy measures the amount of uncertainty or disorder in the data. A good split reduces impurity or entropy and creates purer groups. Gini is generally computationally simpler, while Entropy is based on information gain.

Ques 4: What is a Confusion Matrix? Explain its components.

Ans: A Confusion Matrix is a table used to evaluate the performance of a classification model by comparing actual classes with predicted classes. It contains four main components: True Positive (TP), where the model correctly predicts the positive class; True Negative (TN), where the model correctly predicts the negative class; False Positive (FP), where the model incorrectly predicts the positive class; and False Negative (FN), where the model incorrectly predicts the negative class. These values are used to calculate important classification metrics.

Ques 5: Define Accuracy, Precision, Recall, and F1-Score. Why are these metrics important?

Ans: Accuracy is the proportion of total predictions that are correct. Precision measures how many of the customers predicted as churners actually churned. Recall measures how many of the actual churners were correctly identified by the model. F1-Score is the harmonic mean of Precision and Recall and provides a balance between them. These metrics are important because accuracy alone may not fully describe a classification model, especially when the classes are imbalanced. The experiment specifically requires Accuracy, Precision, Recall, and F1-Score for evaluation.

Ques 6: What is overfitting in a Decision Tree? How can it be reduced?

Ans: Overfitting occurs when a Decision Tree learns the training data too closely, including noise and minor patterns, and therefore performs poorly on unseen data. A very deep tree is more likely to overfit. Overfitting can be reduced by limiting the maximum tree depth, increasing the minimum number of samples required for

splitting or leaf nodes, pruning the tree, or using suitable validation techniques. In the practical implementation, setting max_depth is one way to control the complexity of the tree.

Ques 7: Why is customer churn prediction important for businesses?

Ans: Customer churn prediction is important because it helps businesses identify customers who are likely to discontinue a service or product. Early identification allows organizations to take preventive actions such as offering discounts, improving service, providing personalized support, or introducing loyalty programs. This can improve customer satisfaction, increase retention, and reduce revenue loss.

Ques 8: What are the advantages and limitations of the Decision Tree algorithm?

Ans: Decision Trees have several advantages. They are easy to understand and interpret, can handle both numerical and categorical features, and can represent decision rules visually. They also require relatively little data preprocessing compared with some other machine learning algorithms. However, Decision Trees can easily overfit the training data, can be sensitive to changes in the dataset, and may produce less stable predictions compared with ensemble methods. Proper control of tree complexity is therefore important.

Ques 9: Mention three real-world applications of Decision Tree Classification other than customer churn prediction.

Ans: Three important applications of Decision Tree Classification are medical diagnosis, where the model can classify patients based on symptoms and medical attributes; credit risk assessment, where customers can be classified according to their likelihood of default; and spam email detection, where emails can be classified as spam or non-spam. Decision Trees can be used wherever decisions need to be made based on a set of input features.

Ques 10: How can the insights obtained from a churn prediction model help organizations improve customer retention and business profitability?

Ans: Insights from a churn prediction model help organizations identify customers who are at high risk of leaving and understand the factors associated with churn. Businesses can use this information to create targeted retention strategies such as personalized offers, loyalty programs, improved customer support, and better service plans. By retaining more customers, organizations can reduce customer acquisition costs, minimize revenue loss, increase customer lifetime value, and improve overall profitability. The experiment specifically requires using model predictions and influential factors to recommend strategies for improving customer retention.